VistaLux Entertainment

2025 Growth Forecast

Possible Models?

Historical / Original Linear Model

$\text{Growth} = \beta_0 + \beta_1\text{Income} + \beta_2\text{Production} + \beta_3\text{Savings} + \beta_4\text{Unemployment} + \epsilon_i$

- Where ϵ_i iid~ $N(0, \sigma^2)$

Our Linear Model → (bring in date)

$\text{Growth} = \beta_0 + \beta_1\text{Income}_i + \beta_2\text{Production}_i + \beta_3\text{Savings}_i + \beta_4\text{Unemployment}_i + \beta_5\text{Quarter}_i + \sum_{j=1}^{K}\gamma_j B_j(\text{Date}_i) + \epsilon_i$

- Where ϵ_i iid~ $N(0, \sigma^2)$
- $B_j(\text{Date}_i)$: The j-th basis function of the B-spline for the Numeric Date.
- γ_j: The coefficients (weights) for each basis
- K: The total number of basis functions, which in R is calculated as length(knots) + degree.

Prophet Model

$y(t) = g(t) + s(t) + h(t) + \sum_{j=1}^{n}\beta_j X_{j,t} + \epsilon_t$

- Where ϵ_i iid~ $N(0, \sigma^2)$
- y(t): %growth at time t
- g(t): Trend
- s(t): Seasonality
- h(t): Holidays
- $\sum_{j=1}^{n}\beta_j X_{j,t}$: External regressors (Income, Production, Savings, Unemployment)

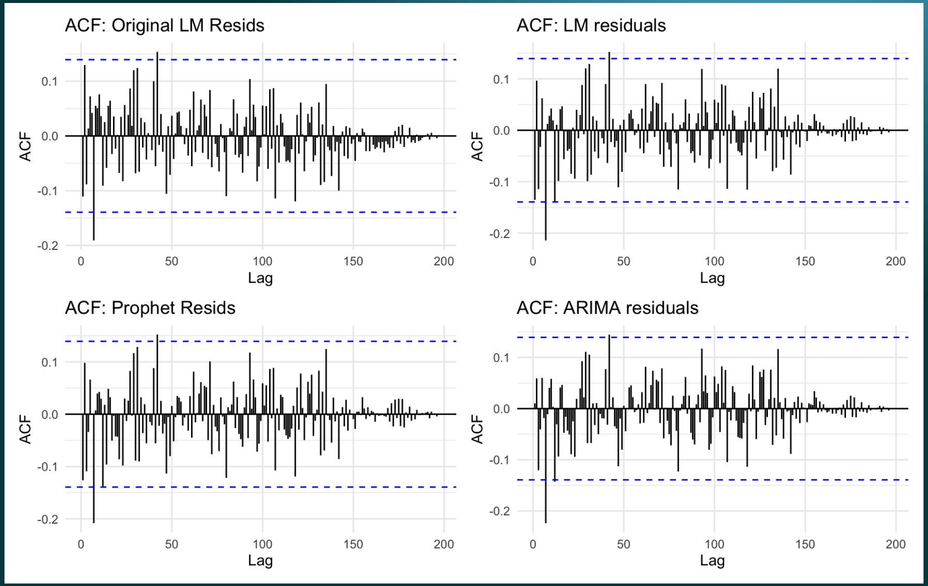
ARIMA Model

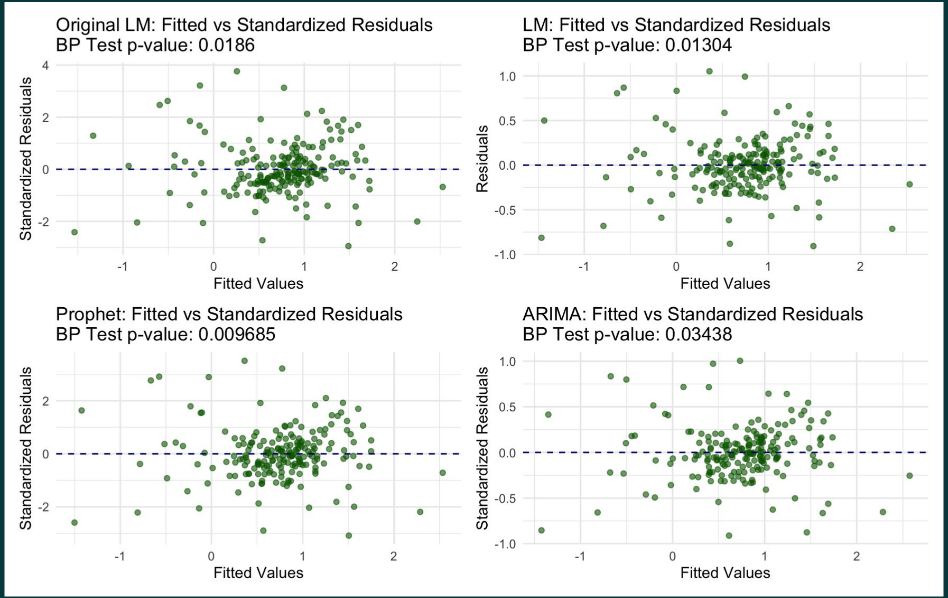
$\text{Growth} = \beta_0 + \beta_1\text{Income}_i + \beta_2\text{Production}_i + \beta_3\text{Savings}_i + \beta_4\text{Unemployment}_i + \beta_5\text{Quarter}_i + \sum_{j=1}^{K}\gamma_j B_j(\text{Date}_i) + \eta_i$

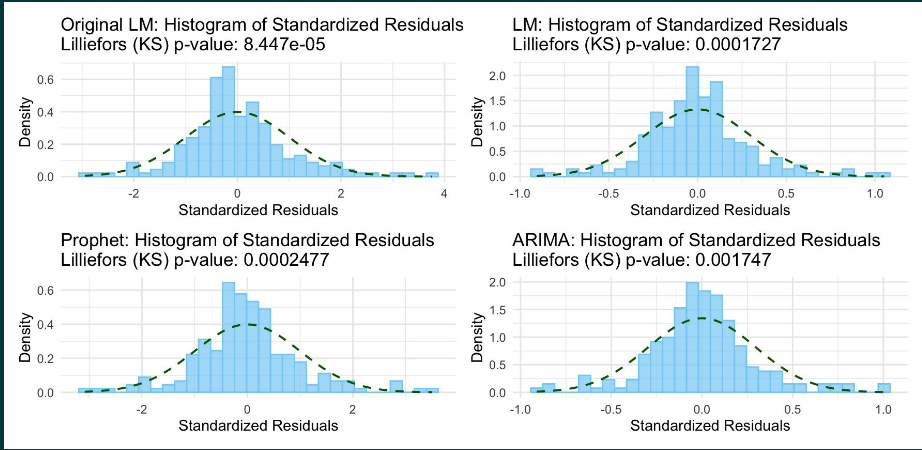
- Where $\eta_i = \varphi\, \eta_{i-1} + \epsilon_i$
- φ : The AR(1) coefficient (autocorrelation parameter). It measures how much the previous residual influences the current one.
- ϵ_i : The "white noise" or "innovation" term, where $\epsilon_i \sim N(0, \sigma^2)$
- $|\varphi| < 1$: The condition for the process to be stationary.

Checking Linearity Assumptions

All 4 models are linear, and there aren’t any obvious patterns, so we continued forward in our analysis.

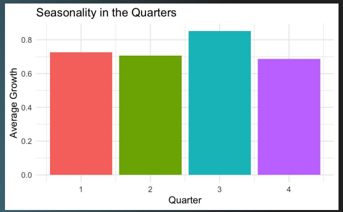






Cross Validation

Model	RMSE
Original LM	0.131
Our LM	0.121
Prophet	0.130
ARIMA	0.123



Historical / Original Linear Model: The original LM model included the regressors income, productions, savings, and unemployment; however, it failed to account for the time series aspect of the data.

Our Multiple Linear Regression Model: Our linear model is different from the original LM model since we accounted for time by adding the predictors Quarter and Date (with cuts in 1988, 2004, 2019). It has the lowest RMSE (meaning it reduces the most error in future predictions).

Prophet Model: We included the regressors income, productions, savings, and unemployment, while using date to model our response variable of percent growth.

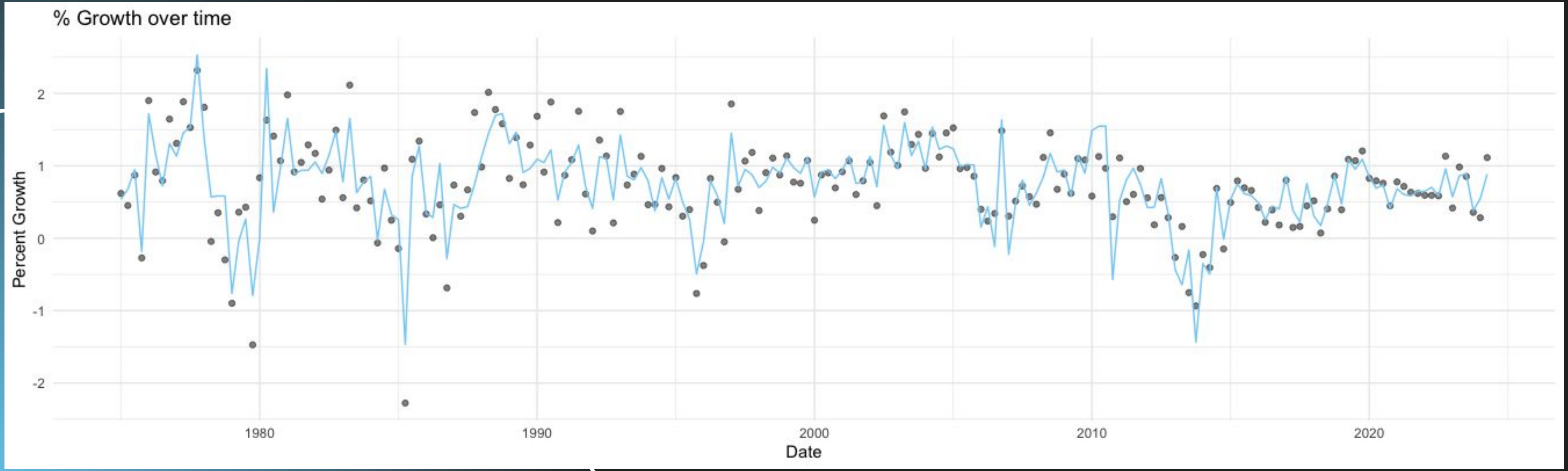
Arima Model: We fit the arima model to try and account for the seasonality in quarters. (ARIMA(1,0,0))

We performed Cross Validation across our four different models (Original, Arima, Prophet, and Linear Model accounting for time) to compare their RMSE and make prediction. Since our Linear Model accounting for dates revealed the smallest RMSE, we proceeded forth with that model for predictions.

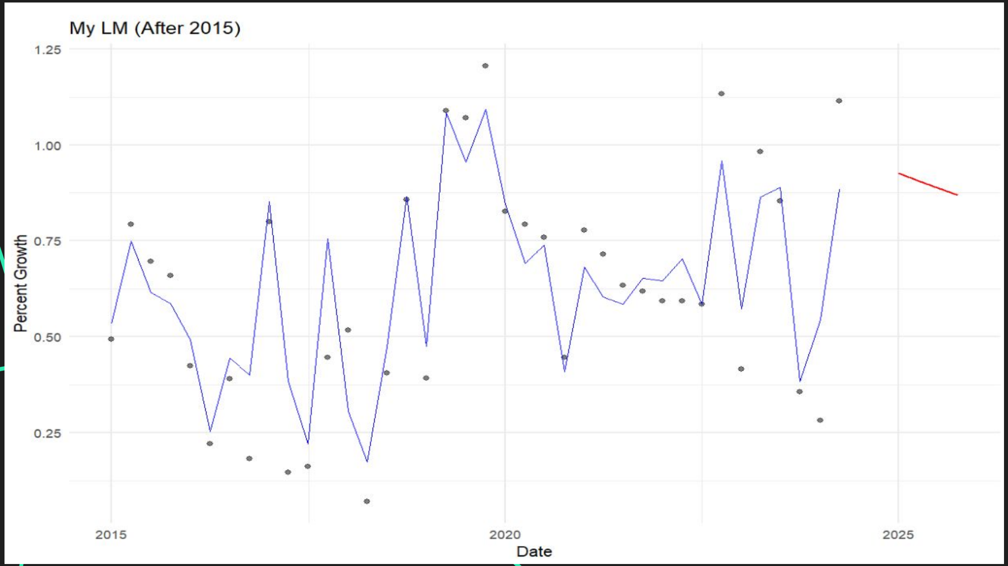
The Model We Chose: Multiple Linear Regression

The current model we decided to use is a Multiple Linear Regression that incorporates both economic indicators (Income, Production, Savings, Unemployment) and temporal dynamics. To capture non-linear trends over time, we used a Cubic B-Spline on the NumericDate variable with specific cut_points (knots). This allows the model to "bend" at certain dates rather than assuming a straight-line growth over years. We also included a Quarter factor to account for fixed seasonal effects.

Performance: This model achieved an RMSE of 0.121, the lowest among all tested candidates, indicating it has the highest predictive accuracy for this specific dataset.



2025 Projections Under Our Linear Model



The red in the graph to the left illustrates the 2025 predictions for our chosen Linear Model. To obtain predictions, we first created a quarterly future dataframe for 2025 using our last observed regressor values(Income, Production, Savings, and Unemployment). We then applied our fitted LM model to obtain the predicted Percent Growth values for each quarter. The historical fitted values from the model are shown in blue, allowing comparison between observed trends and projected growth. Based on the numbers we got, we predict a decrease in Percent Growth during 2025.

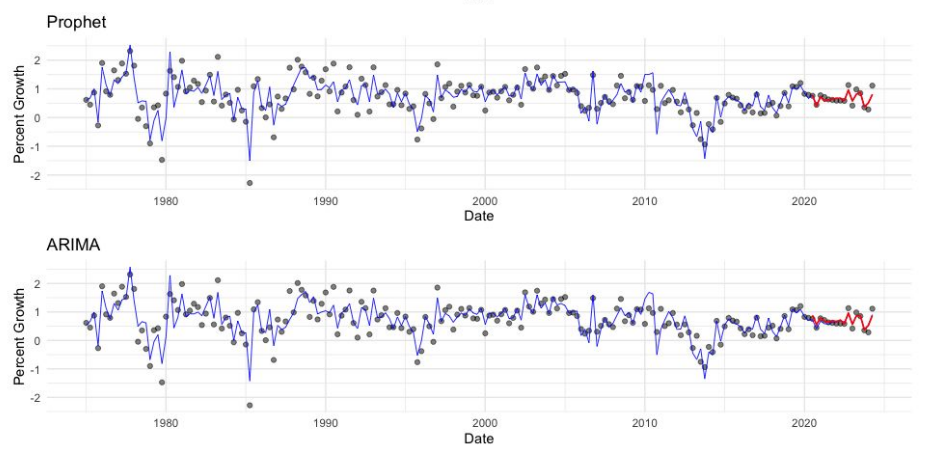
The Models We Didn't Choose: Prophet & ARIMA

Alternative A: Prophet (Additive Model)

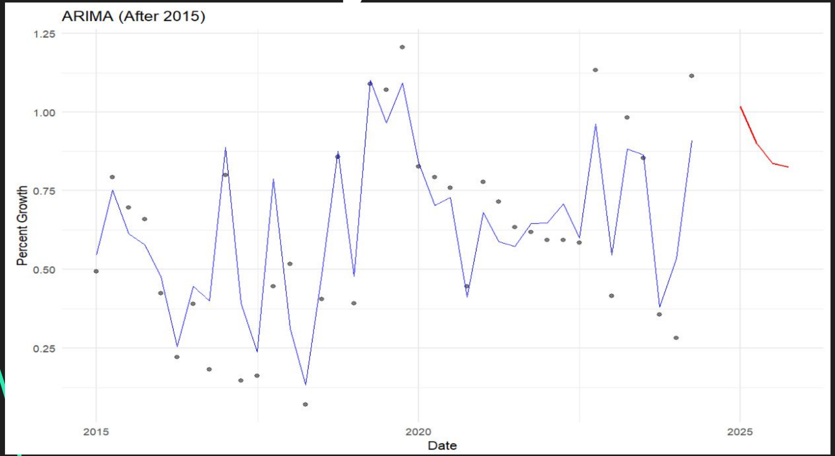
- Unlike the LM, which treats time as a series of spline basis functions, Prophet decomposes the time series into a structural trend ($g(t)$) and periodic seasonality ($s(t)$). While it also accepts external regressors, it is designed to handle "changepoints" and holiday effects more natively than a standard linear model, meaning that without more granular date data, Prophet can't perform to its maximum potential, explaining the RMSE of 0.13.

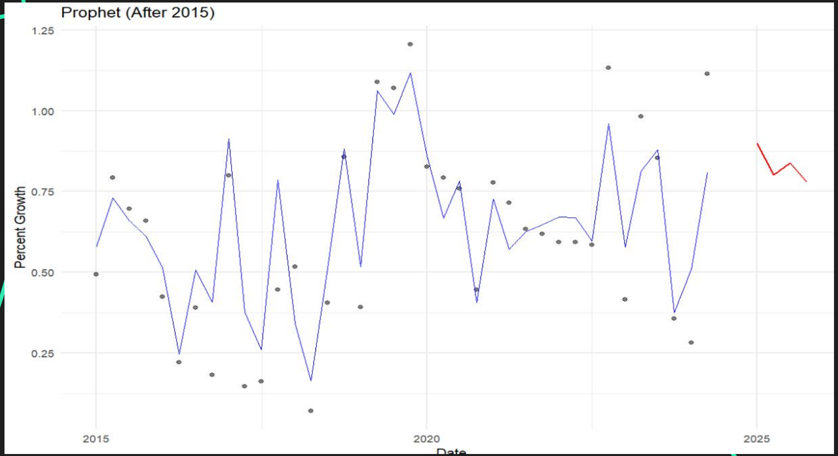
Alternative B: Regression with ARIMA(1,0,0)

- While the LM assumes that residuals are independent and identically distributed (i.e., no correlation between today's error and yesterday's), the ARIMA(1,0,0) model explicitly accounts for autocorrelation. It assumes that deviations from the trend persist over time (which does make sense for time series data, and likely why it performed close to our Multiple Linear Regression Model with an RMSE of 0.123).



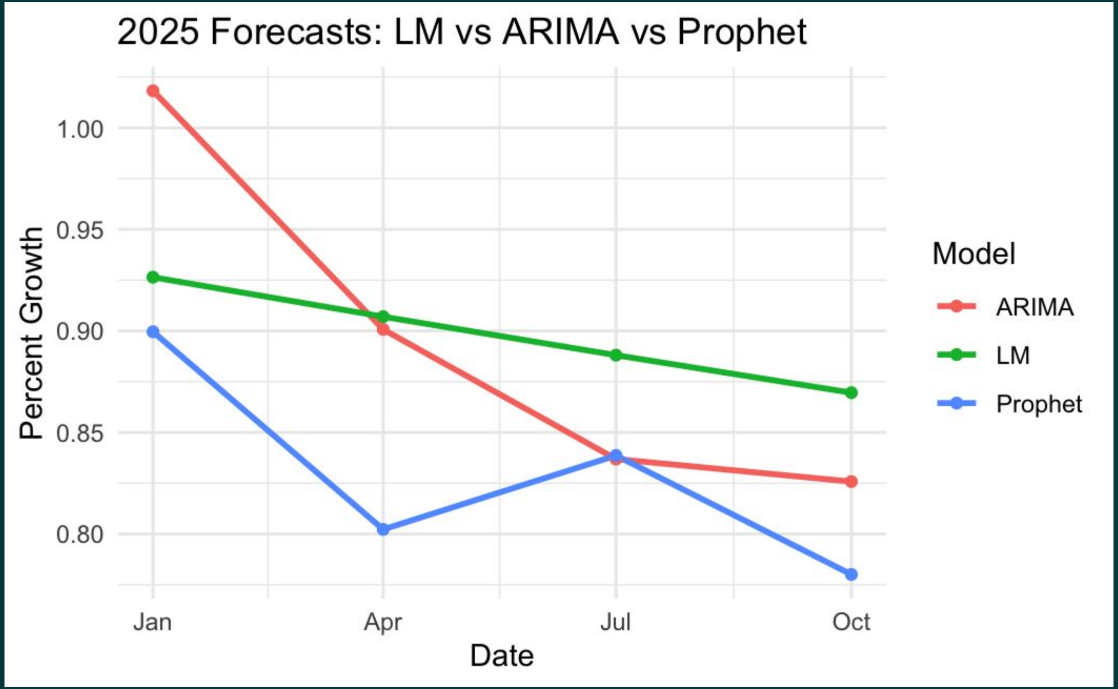
2025 Projections Under Other Models





As we saw from our RMSE values earlier, we know that ARIMA and Prophet don't perform quite as well at prediction as our Linear Model. The same approach from finding our 2025 LM predictions (as seen in the slide prior) was applied to generate 2025 forecasts using our Prophet and ARIMA models.

Comparing Model 2025 Predictions



From this graph and comparison we know that regardless of which model chosen, one can expect to see a decreasing trend for Percent Growth in 2025.

Income, Production, Savings and Unemployment (from Our Linear Model)

Regressor	Beta Coefficient
Income	0.7280
Production	0.0314
Savings	-0.0522
Unemployment	-0.2523

By looking at the coefficients for each factor, we can see how they impact Percent Growth. For example, holding all else constant, as income increase by one unit, then we can expect a 0.728 increase in Percent Growth for VistaLux.

Income → How much consumer spending power drives our growth.

Production → The impact of broader industrial output on our performance.

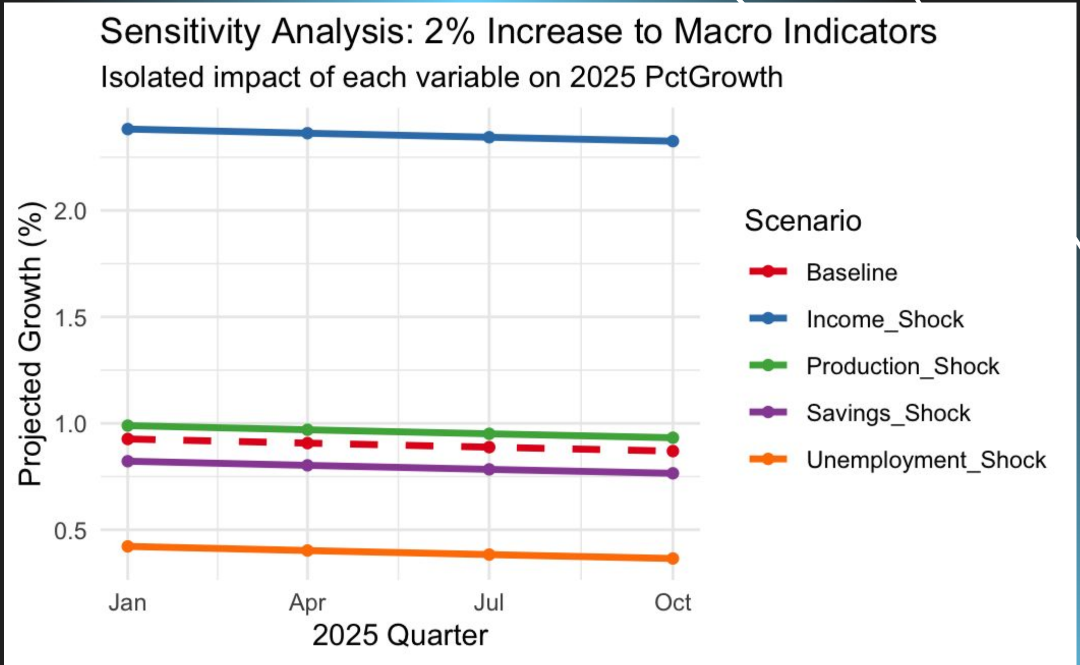
Savings → Whether a "thrifty" consumer (high savings) helps or hurts our revenue.

Unemployment → Our sensitivity to labor market shifts and job security.

Income, Production, Savings & Unemployment in Certain Scenarios

Using our Linear model we looked at predictions for 2025 increasing each regressor by 2% independently and seeing how the predictions were impacted as compared to the Baseline prediction (same as the prediction we looked at before, just on a wider scale).

An increase in Income, again, seems to show the best increase in predicted percent growth for our company, while an increase in production slightly helps, an increase in penny-pinching by the populace slightly decreases our percent growth, and unsurprisingly, an increase in unemployment rates decreases our growth.



Thank you!